Oscar I. Hernandez

Summary

I'm an incoming graduate student of the mathematical sciences.

Email: oscar.hernandez@simons-rock.edu Website: https://mathemonads.github.io
Phone: +1 (646) 961-3715 GitHub: http://github.com/mathemonads

Education

Bard College at Simon's Rock (BCSR) Aug 2013 – May 2017
BA, Mathematics and Computer Science, Honors Thesis [HW17] – May 2017
AA, Liberal Arts and Sciences Aug 2013 – May 2015

Awards: W.E.B. Du Bois Scholarship (2013–2017), Alumni Scholarship (2013–2017), Supplemental Return Grant (2014–2017), Dean's List (Fall 2015 and Fall 2016).

Experience

Freelance Software Engineer	Independent Contractor	Jan 2018 –
Founder & Robotics Engineer	ARKS	Apr 2018 – Jan 2020
Research Assistant	BCSR: Economics	Spring 2016
Research Intern	BCSR: Computer Science	Summers 2015 & 2016

Teaching

5th Grade Humanities Teacher	SACS HNW	Jul 2017 – Dec 2017
G5 World History	G5 Math	G5–6 CS Quarter Elective
G5 English Language Arts	G5 Earth Science	G5–8 CS Semester Elective
Subject/Class Tutor	BCSR	Fall 2014 – Spring 2017
Computer Science	Latin I	Intro to Computer Science
Critical Reasoning	Formal Logic	Algorithms & Data Structures
Algebra I with Geometry	Discrete Math	Graph Theory & Combinatorics
Algebra II and Trigonometry	Vector Calculus	

Awards

1. Intel Edge AI Scholarship (Udacity 2019)
2. Machine Learning with Tensorflow on GCP Specialization (Coursera 2020)
3. NSF I-Corps grant awardee x2 (UCR Spring 2019, CSUN Fall 2019)
4. Best Social/Civic Hack (Citrus Hack 2018), and
   Superstar Award & Effective Use of Technology Award (Hack SRC 2017).

Publications

[Her+18] Oscar I. Hernandez et al. "Constructing de Bruijn sequences with co-lexicographic order: The k-ary Grandmama sequence". In: *European Journal of Combinatorics* 72 (2018), pp. 1–11. ISSN: 0195-6698. URL: http://www.sciencedirect.com/science/article/pii/S0195669818300696.

[Has+17] Harold M. Hastings et al. "Dynamics of Biomimetic Electronic Artificial Neural Networks". In: *Proceedings of the 4th International Conference on Applications in Nonlinear Dynamics (ICAND 2016)*. Ed. by Visarath In, Patrick Longhini, and Antonio Palacios. Cham: Springer International Publishing, 2017, pp. 195–207. ISBN: 978-3-319-52621-8. URL: https://link.springer.com/chapter/10.1007%5C%2F978-3-319-52621-8_18.

[HW17] Oscar I Hernandez and Aaron Williams. "Constructing Maximum-Rotating de Bruijn Sequences using Necklace Trees". *Bachelor of Arts* Senior Thesis awarded Honors. Bard College at Simon's Rock: Division of Science, Mathematics, and Computing, May 2017. URL: https://digitalcommons.bard.edu/sr-theses/1120.

[DHW16] Patrick Baxter Dragon, Oscar I. Hernandez, and Aaron Williams. "The Grandmama de Bruijn Sequence for Binary Strings". In: *LATIN 2016: Theoretical Informatics*. Ed. by Evangelos Kranakis, Gonzalo Navarro, and Edgar Chávez. Berlin, Heidelberg: Springer Berlin Heidelberg, 2016, pp. 347–361. ISBN: 978-3-662-49529-2. URL: https://link.springer.com/chapter/10.1007/978-3-662-49529-2_26.

Workshop papers, invited presentations, and small articles:

[HI19] *Zero Shot Sentiment Analysis on Tweets in Any Language*. Poster Session #1. Southern California Natural Language Processing Symposium. University of Southern California, Sept. 2019. URL: https://socalnlp.github.io/symp19/index.html.

[HT19] *Toward Compositional Transfer Learning*. Contributed Paper Session 1. American Mathematical Society Fall Western Sectional Meeting. UC Riverside, Nov. 2019. URL: http://www.ams.org/amsmtgs/2266_abstracts/1153-18-420.pdf.

[Her+19] *Spinning Around and Around: Turnstile Puzzles are PSPACE-Complete*. Work under revision started at the invite-only 32nd Bellairs Winter Workshop on Computational Geometry. 2019.

[HY18] *Automating Art Snobbiness*. Special Interest Group on Harry Q Bovik. Similar joke articles published annually since 2016. Association for Computational Heresy. Carnegie Mellon University, Apr. 2018. URL: http://sigbovik.org/2018/proceedings.pdf#page=94.

[Her+16] *Analog Artificial Neural Networks*. Poster Session. Awarded 2nd place and $200 https://www.aps.org/units/nyss/governance/minutes/upload/Fall16.pdf#page=4. New York State Section of the American Physical Society (NYSS-APS) Symposium on the Physics of 2D Materials. Rensselaer Polytechnic Institute, Nov. 2016. URL: https://simons-rock.edu/news/students-win-prize-physics-research.php.